

Final Design Presentation

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Problem & Function Overview - Design Problem & Strategy

Launch & Deploy: Leave Dragon Stables.

Our strategy: Dual-axle drivetrain for fast, stable deployment.

Extinguish Fire: Hit lever (~42").

Our strategy: 3-Stage Telescoping Arm made of PVC pipes

Refill Feeding Stations: Collect fish; avoid eels (or bin eels).

Our strategy: Chose to neglect due to time constraints & ensure other mechanism worked $\leq 95\%$

Return Sheep: One sheep per paddock (no duplicates).

Our strategy: Pneumatic piston “sheep shooter” for controlled launches.

Light Bonfire: Pull torches to target rings.

Our strategy: Dual-link falling “wing” arms for sweeping torch motion onto the target.

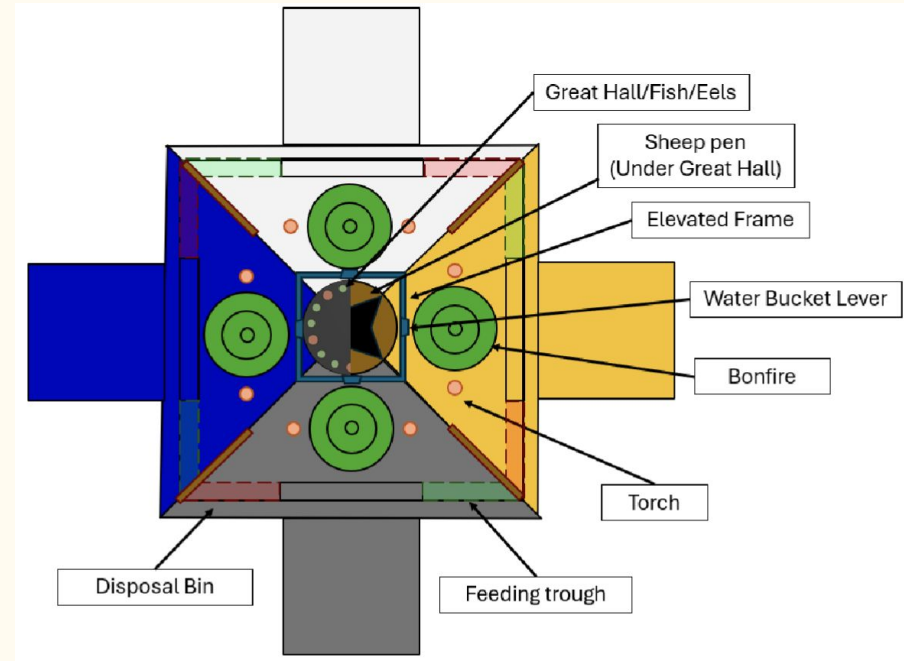
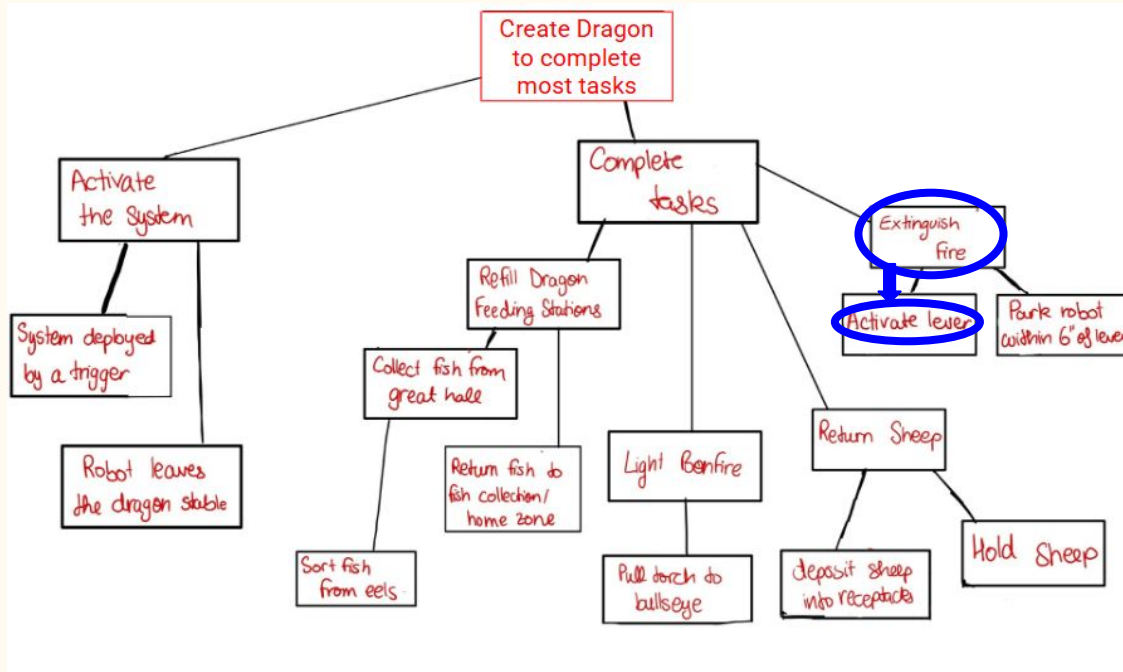


Figure 1. ME 2110 “Train Your Dragon” competition arena layout (adapted from course materials, Georgia Tech ME 2110, 2025).

Problem & Function Overview - Function Tree

Throughout development we adopted a iterative design & fabrication strategy through CAD modeling & simulation, and extensive prototyping & physical testing.

However, in the earlier stages of our project, the effective use of morph charts and a function tree, where crucial in streamlining concept selection and identifying the most effective mechanisms within our various constraints.

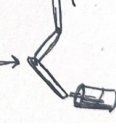
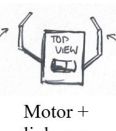
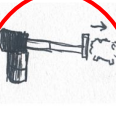
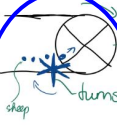


The function tree specifically was essential in the early design phase, helping us define each task in engineering terms.

Instead of “Extinguish Fire”, we reframed it as “Activate Lever”, clarifying what the robot must physically accomplish.

Problem & Function Overview - Morph Chart(s)

Similarly, the morph chart helped guide our design stage after the function tree. While the function tree clarified each task physically, the morph chart helped us visualize and compare multiple mechanisms, leading our team to select 1–2 final concepts to CAD and fabricate.

Extinguishing Fire	 Linear Actuator and wood piece	 Motor powered lever	 <u>Motor with 3 bar linkage</u>	 Mousetrap with rubber band
Lighting Bonfire	 <u>Motor with lever arms</u>	 Linear actuator and tape	 Motor + linkages	
Returning Sheep	 Motion Sensor to detect sheep	 Linear actuator to push sheep	 Conveyor belt	 <u>Turnstile + Ramp</u>

For example...

The morph chart led us to two designs for returning sheep:

A simple mechanical turnstile dispenser, which is easy to build but less reliable. And a pressurized pneumatic piston system, which is more complex but far more consistent.

We CAD-modeled, simulated, physically prototyped, and tested both, ultimately helping us select the piston shooter as our final mechanism.

This is only a portion of the full morph chart.



Subsystem-Based Design Overview

This design includes...

- ❖ A motor powered drive train
- ❖ A switch-activated dispenser and piston system
- ❖ A hinged sweeping mechanism released by a bar
- ❖ Three-piece telescoping arm

Drive Train

- A series of gears and axles transfer the rotation of the motor to the rear wheels
- Gears are connected by drive belts

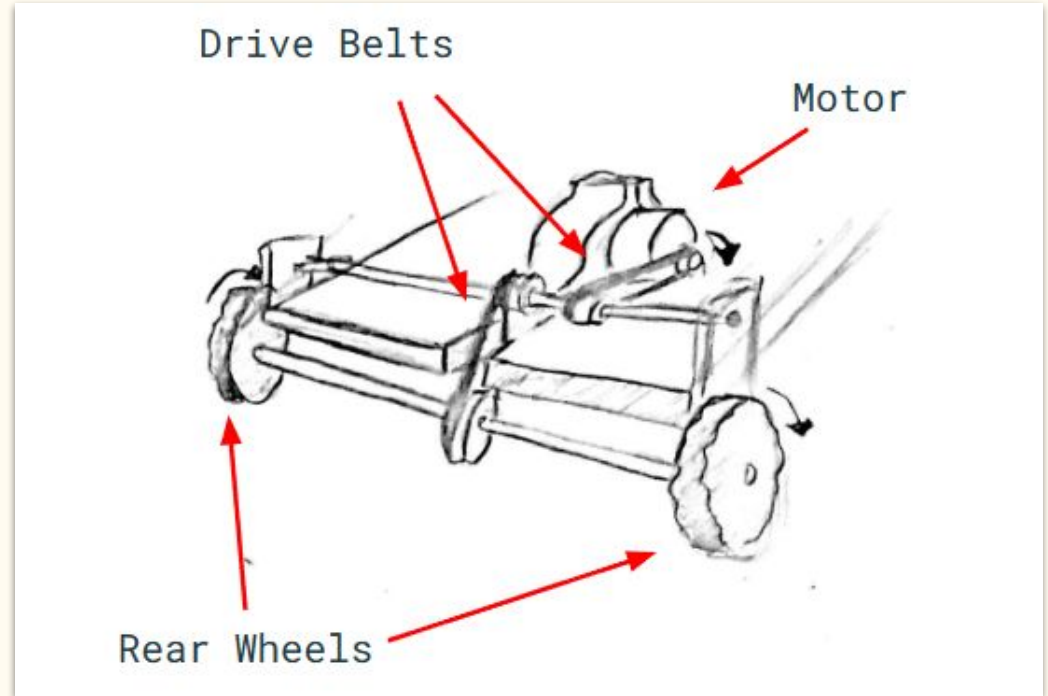


Diagram showing placement of gears and axles

Drivetrain Iteration 1

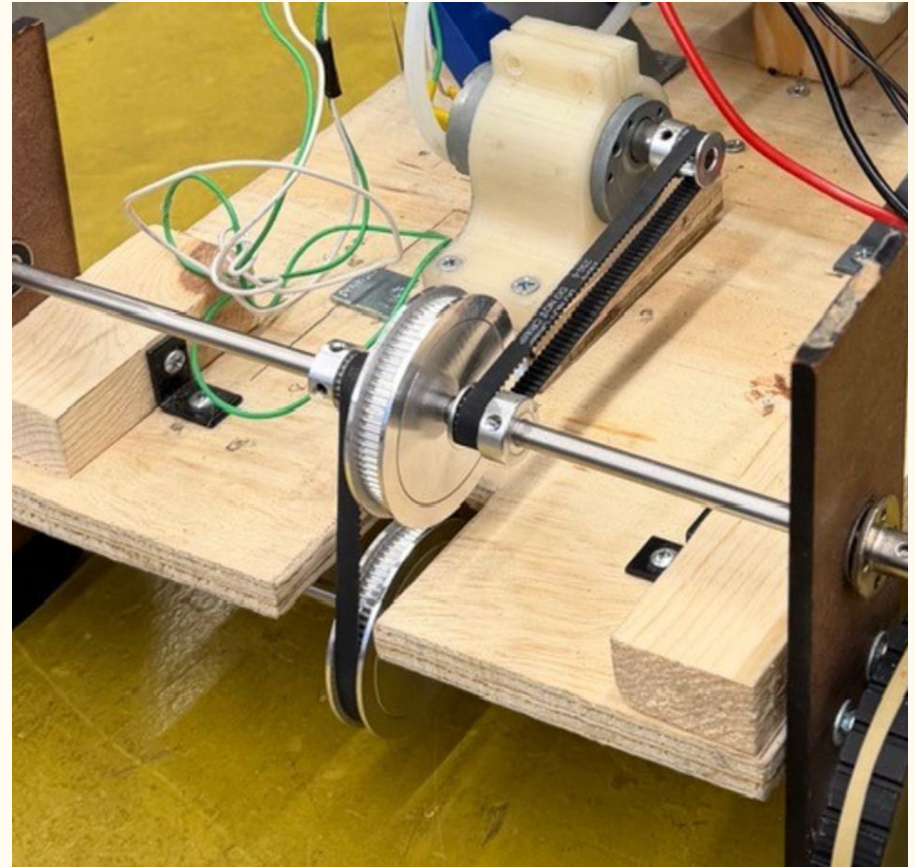
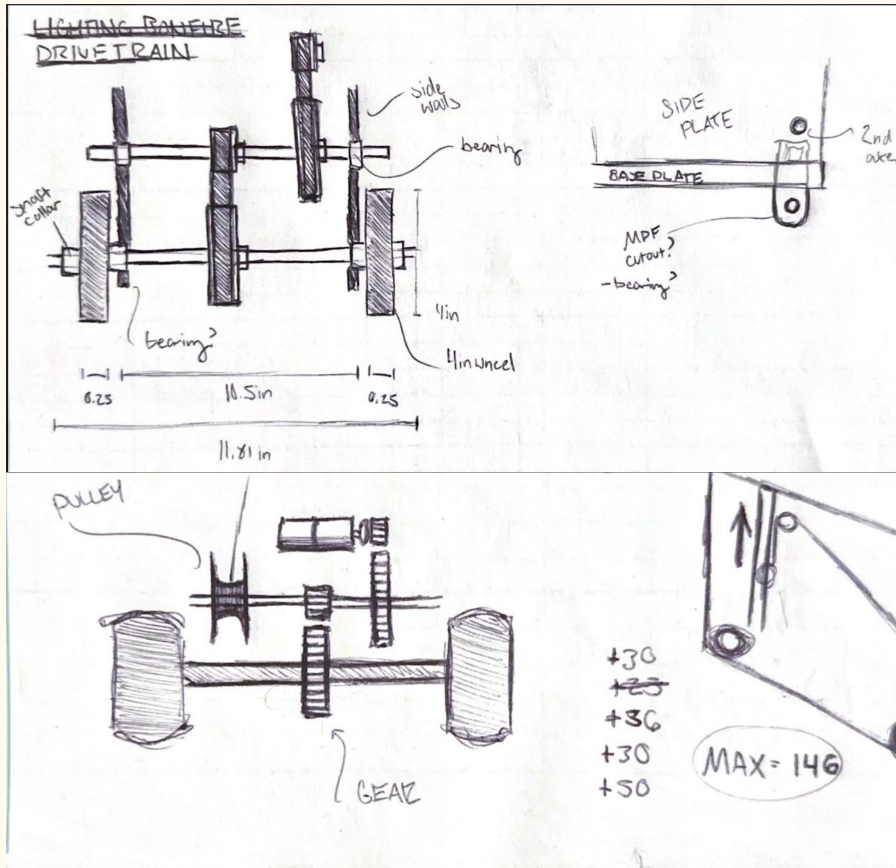
Metric: Drives robot for a specified amount of time 100% of the time

Dual-axle drivetrain using conveyor belts and gears.

- **Testing:** Measured time to traverse field and alignment drift.
- **Observation:** Performance affected by wiring drag and belt tension. If wires get tangled/dragged, wheels did not have enough friction to keep moving robot.
- **Root Cause:** Wire routing impacted balance and wheel contact.
- **Design Evolution:** Re-routed wires, improved belt tensioning, and added rubber bands to wheels to improve friction.

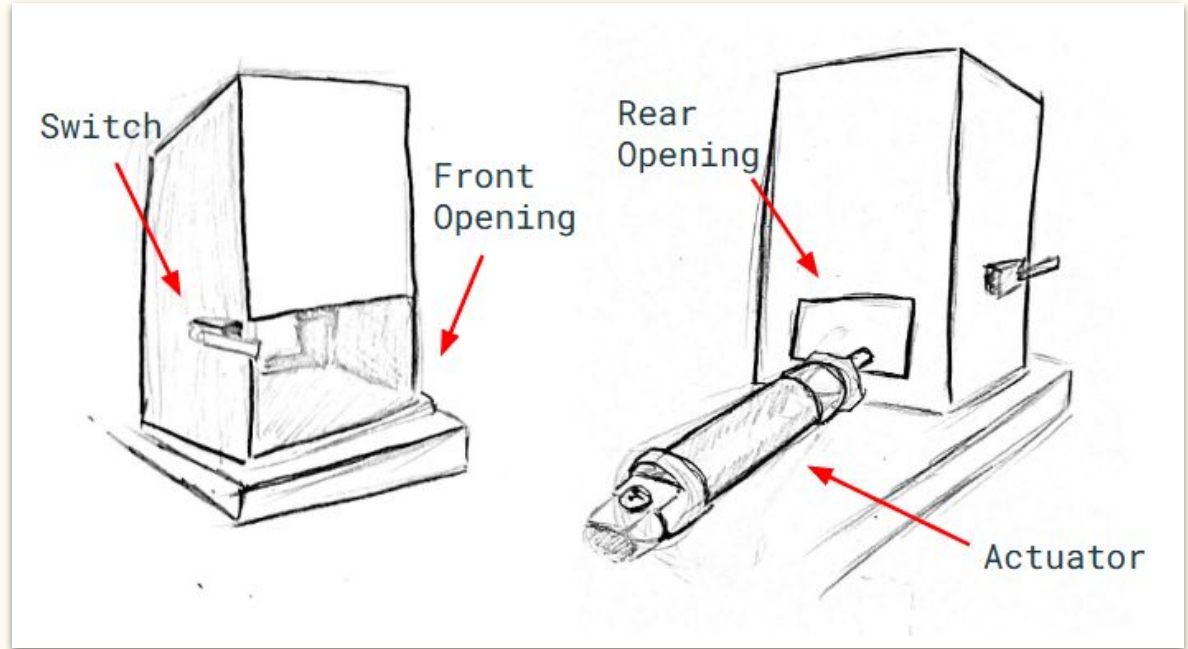


Drive Train



Dispenser-Piston System

- Dispenser is mounted to a board with the actuator mounted behind it
- Switch is mounted on side of dispenser so that the piston is activated when it hits the walls of Great Hall



Front of dispenser, opening to allow for sheep propulsion.

Back of dispenser, actuator cylinder aligned with rear opening.

Sheep Subsystem Iteration 1

Metric: 95% successful sheep deployment

Turnstile with Ramp

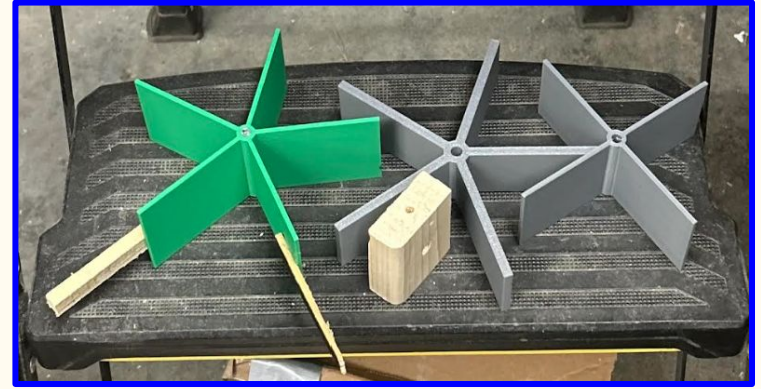
- **Goal:** Successfully deploy sheep into receptacles
- **Testing:** Varying ramp angles, arm lengths, and wall distances.
- **Observation:** Sheep frequently jammed(85% of the time). Sheep don't slide down ramp.
- **Root Cause:** Surface friction and distance from wall is too small



Sheep Subsystem Iteration 2

Metric: 95% successful sheep deployment

Turnstile with Vertical Dispenser



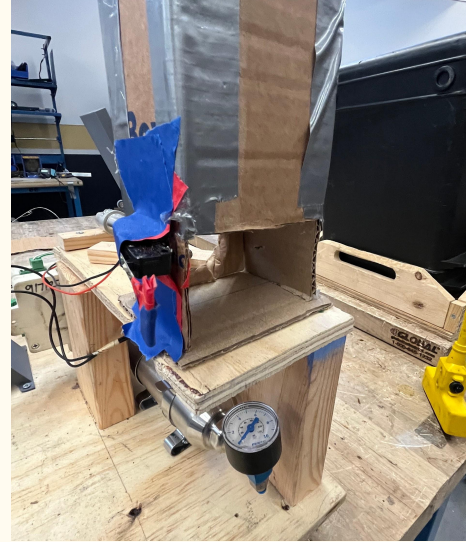
- **Goal:** Reduce friction and improve feed alignment.
- **Testing:** Varied dispenser widths and arm lengths
- **Observation:** Jamming reduced to 40%, but still persistent
- **Root Cause:** Inconsistent feed path

Sheep Subsystem Iteration 3

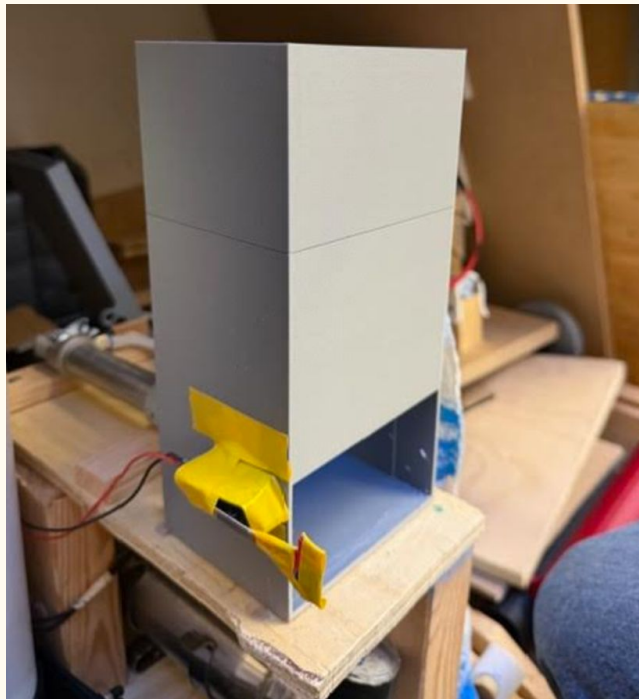
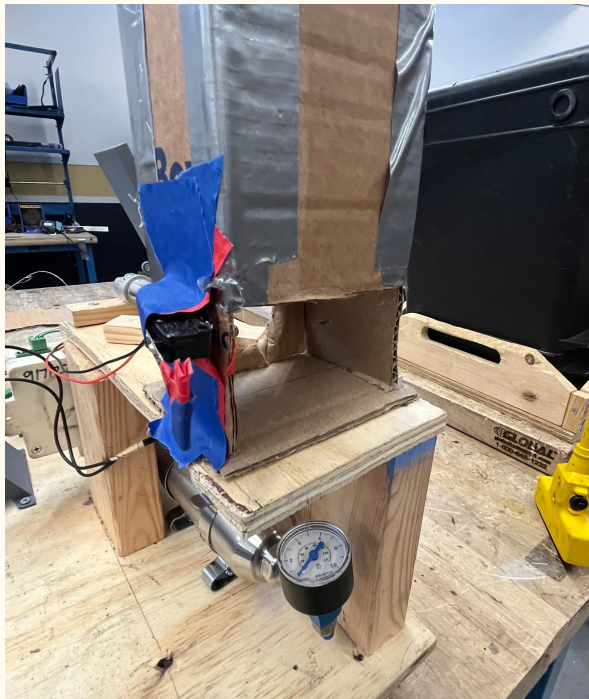
Metric: 95% successful sheep deployment

Piston-Actuated Vertical Dispenser

- **Goal:** Achieve reliable one-sheep release per activation
- **Testing:** Varied dispenser widths, position of switch, timing of piston activation
- **Observation:** ~95% successful sheep deployment. ~5% of times 3rd/4th sheep falls on its side, which causes it to jam and not deploy.
- **Root Cause:** Length/width of dispenser is too big, allowing sheep to rotate as they falls
- **Fix:** Adjusted length and width of vertical dispenser

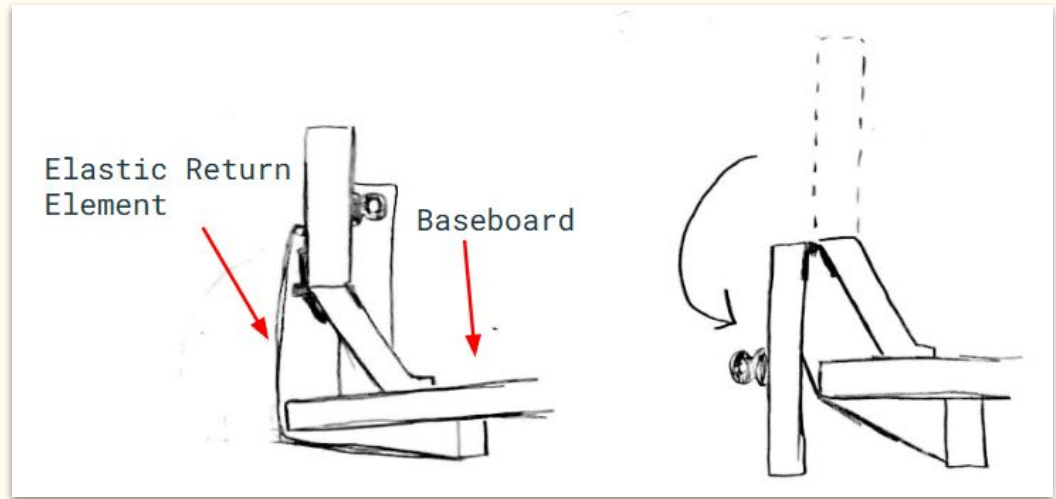


Sheep Dispenser



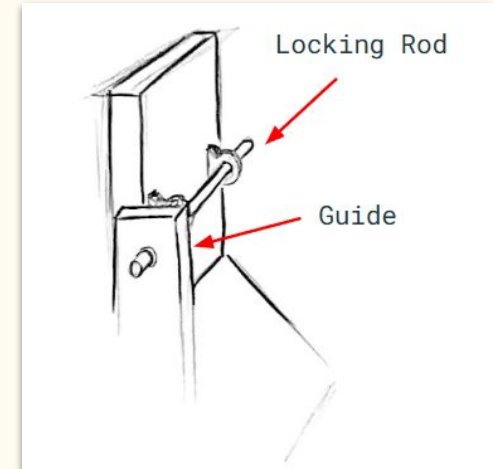
Sweeping Mechanism

- Two hinged arms mounted on either side of the baseboard.
- Held in the raised position by a locking rod inserted through a guide.
- Top linkages are connected to the bottom of the baseboard with a rubber band.
- Once rod is removed, the tension rotates the top arm downwards, sweeping the torches towards the target.



Sweeping mechanism held in raised and released state (top).

Diagram showing locking rod position (bottom right).



Lighting Bonfire Iteration 1

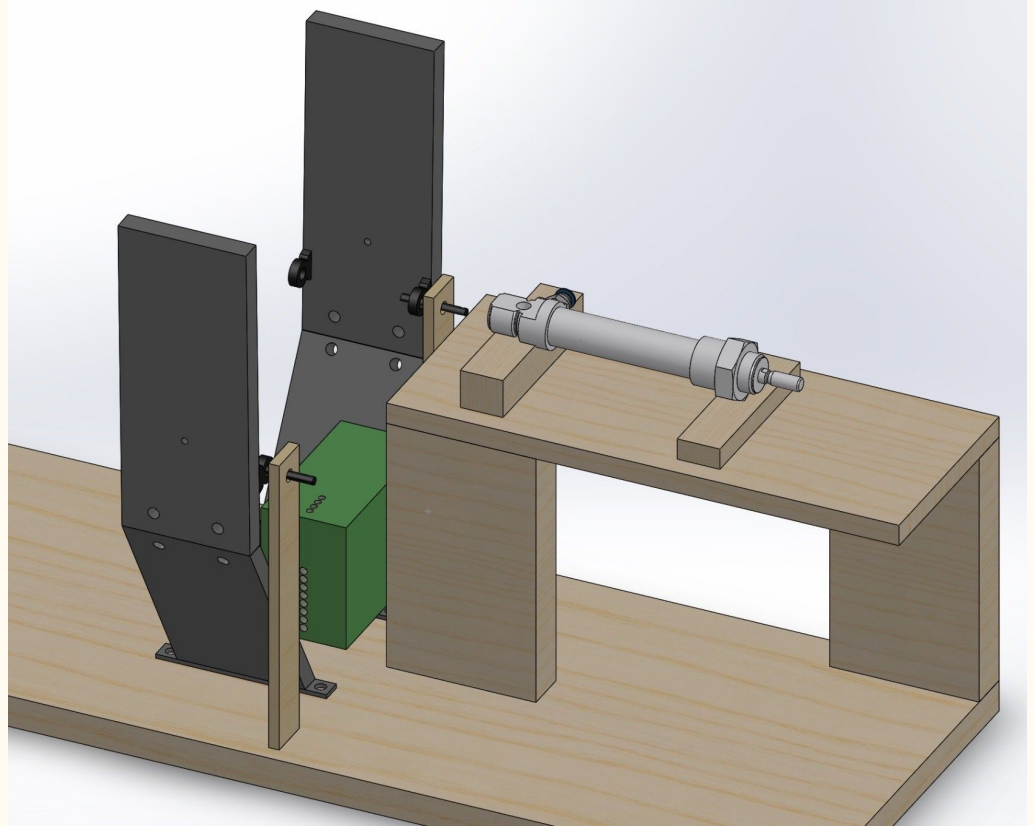
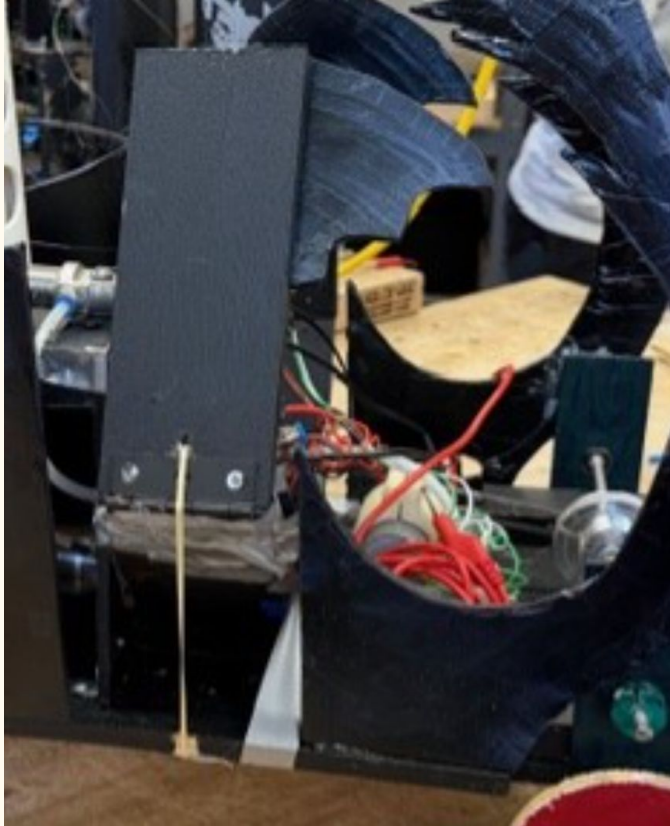
Metric: Torches touch 2nd ring 95% of time or bullseye 70% of time

“Wing” Arms

- **Goal:** Simultaneously pull torches toward target bullseye.
- **Testing:** Tried various rubber band lengths to adjust stored spring energy, adjusted length of string for the trigger
- **Observation:** String can shift and arms can be deployed too early. 55% of time, arms deployed at the wrong time
- **Root Cause:** String can shift on the axel and is hard to attach so its static.
- **Improvements:** Move string from axle to piston, which will activate when turnstile rotates



Sweeping Mechanism



- Concentric pipe system made up of three PVC pipes with a nylon thread connecting the bottom of each inner pipe with the top of each outer pipe.
- Stoppers are attached to the bottom of both inner pipes and the tops of each outer pipe so pipes extend straight. without falling out
- Nylon thread attached to spool on a motor pulling the inner pipes up

Telescoping Arm

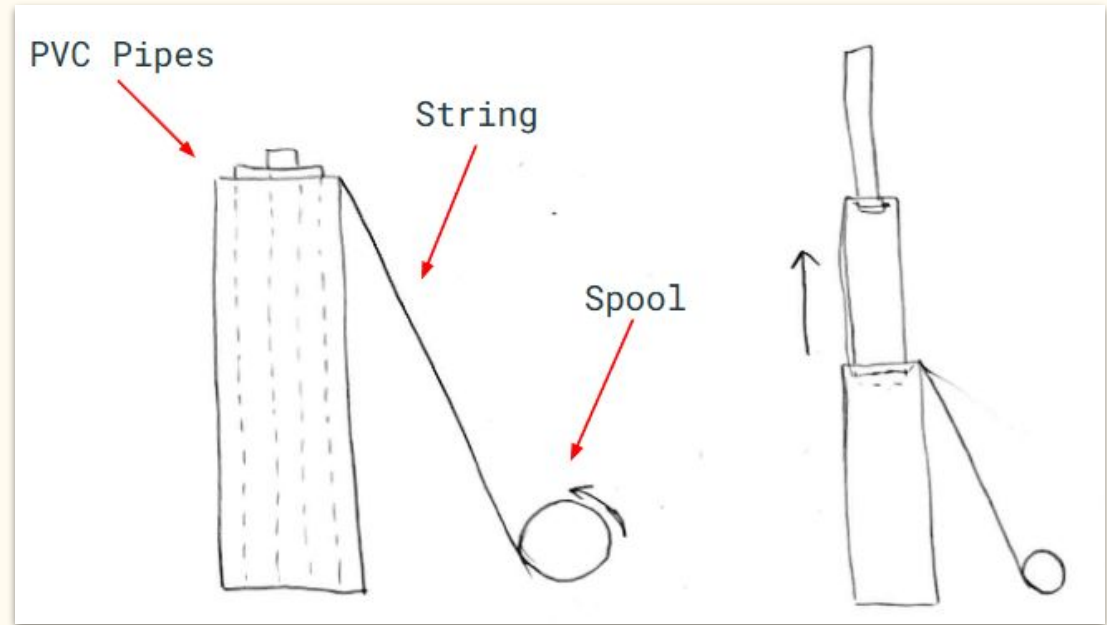


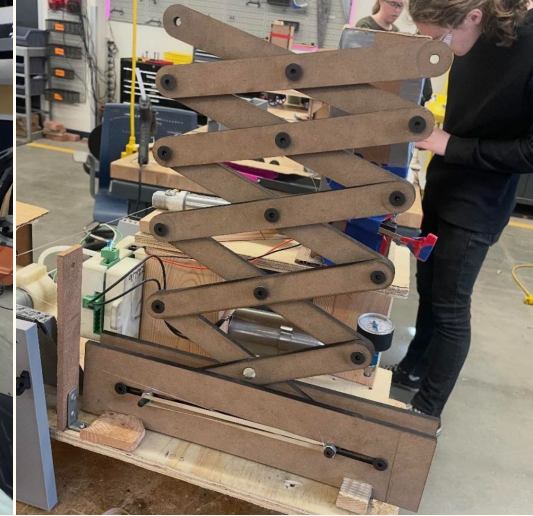
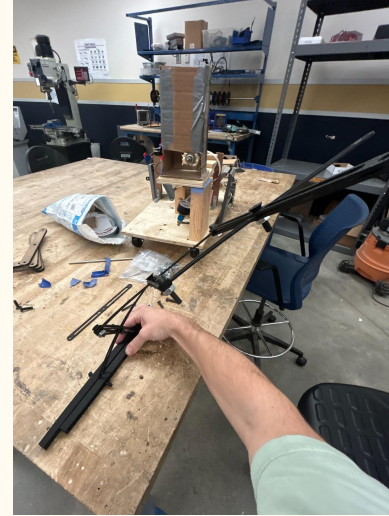
Diagram showing telescoping arm in resting and raised state

Extinguish Fire Iteration 1

Metric: Activate lever 95% of the time and be within 6" of it 100% of the time

Scissor Arm

- **Goal:** Extend arm to flip water bucket
- **Testing:** Tested different material and activation mechanisms
- **Observation:** (MDF) Motor did not have enough torque to pull the scissor arm up (3D printed) Leaned too much
- **Root Cause:** MDF was too heavy, PLA was too thin and flimsy
- **Improvements:** - Switched from MDF to PLA; Stacked 2 arms to improve lean



Extinguish Fire Iteration 2

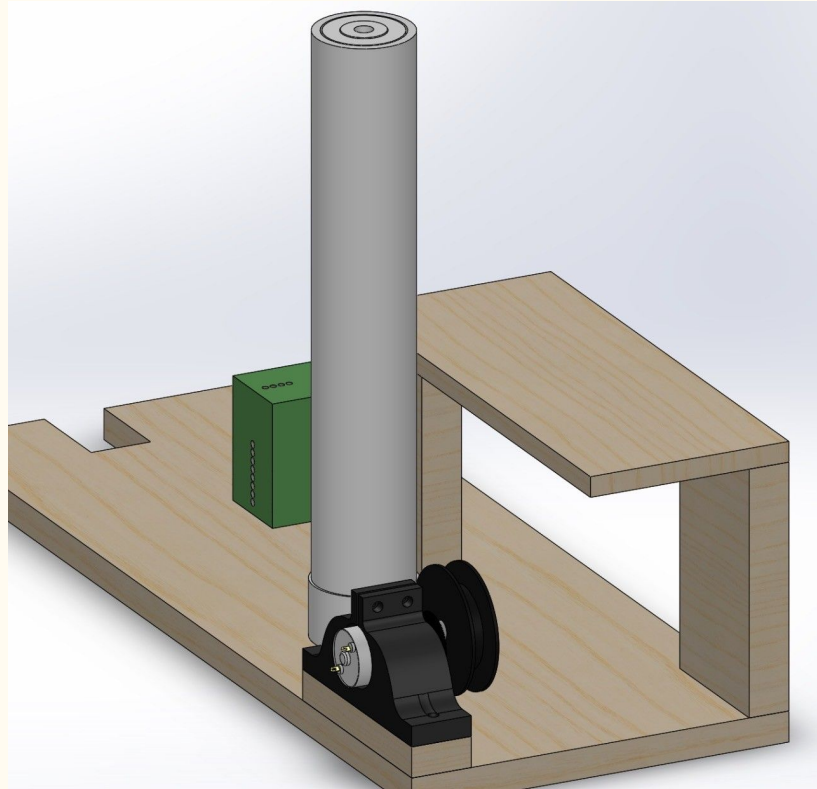
Metric: Activate lever 95% of the time and be within 6” of it 100% of the time

Telescoping Arm

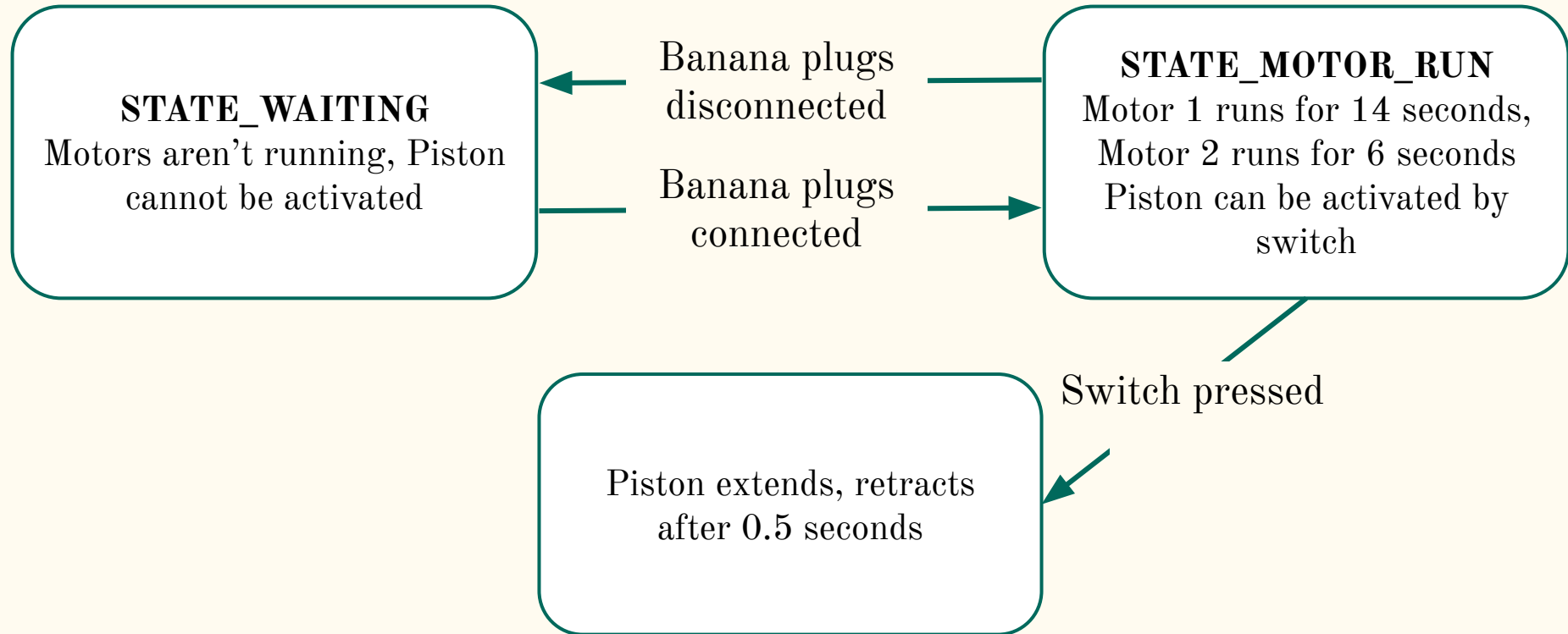
- **Goal:** Extend arm to flip water bucket
- **Testing:** Tested different size PVCs, tolerances, and material
- **Observation:** Heavy and inconsistent lean, motor needs lot of torque to lift up arm, missed the water bucket during sprint due to inconsistencies
- **Root Cause:** Tolerance on 3D print too high, cardboard loosens over time
- **Improvements:** 3D print toppers, use smaller diameter PVC to reduce weight
- **Activated lever 98% of the time after all improvements**



Extinguish Fire Iteration 3



Control and Sensing Flowchart



Sprint Results (1 & 2)

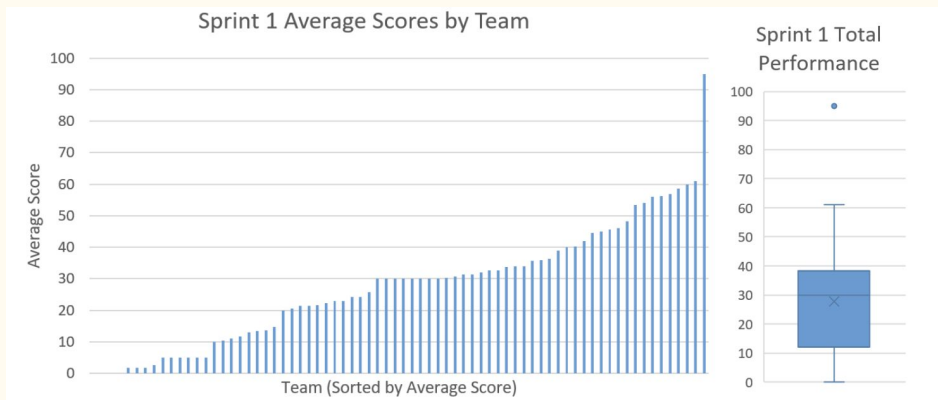
Sprint 1

Median: ~30 points

Our score: 30 points

Next steps: Continue designing, CADing, & simulating/testing objective mechanisms

Begin fabrication of Return Sheep & Light Bonfire

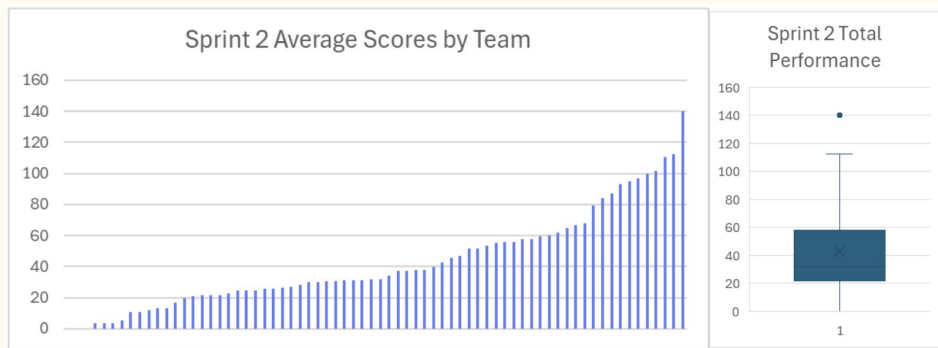


Sprint 2

Median: ~40 points

Our score: ~50-70 points

Next Steps: Finalize Return Sheep & Light Bonfire mechanisms to desired success rate ($\geq 95\%$), continue iterative testing for extinguishing fire mechanism



Post Sprint 2:

After Sprint 2, focus shifted from functional prototypes to finalized designs that were more precise, durable, and aesthetically pleasing.

BY SPRINT 3: 90% completion rate for the extinguish fire task using a scissor arm/ telescoping prototype. Finalized designs for the sheep dispenser and sweeping mechanism.



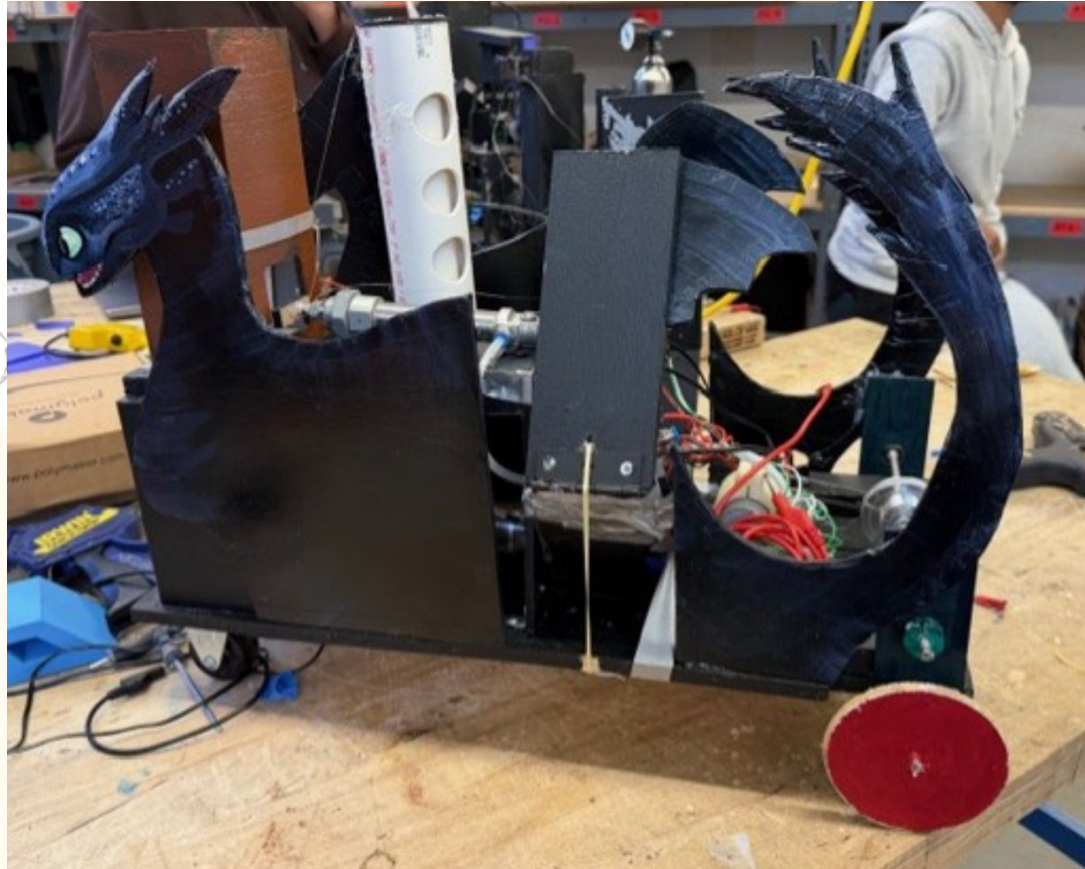
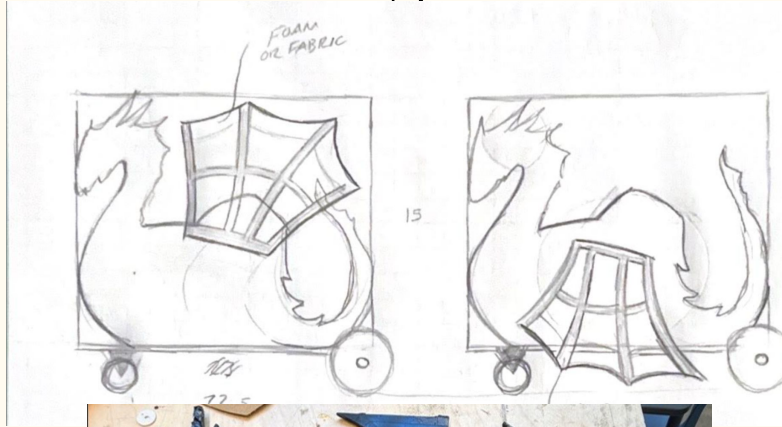
BY FINAL COMP WEEK: 95%+ completion rate for all tasks attempted. Finalized design for the extinguish fire task.

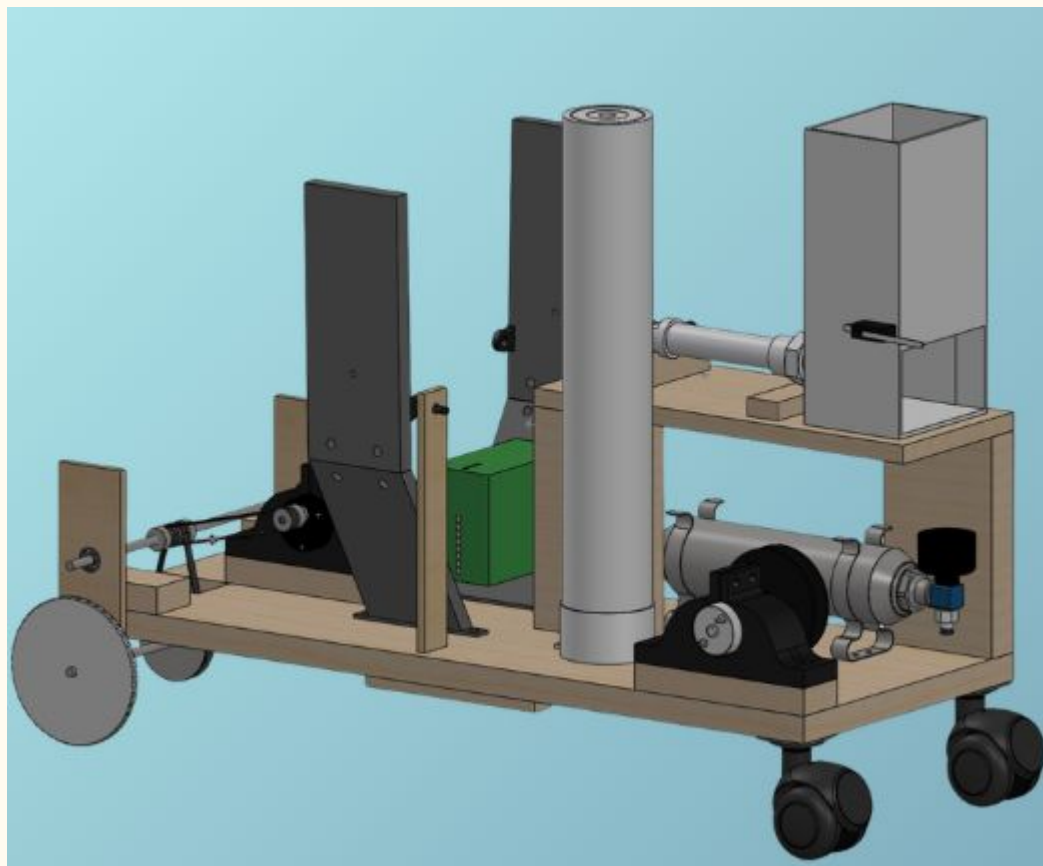


Completion rate is based upon average points scored divided by the maximum amount of points to be scored, based on the tasks being attempted.

Our max: **146 points**

Final Design





Subsystem	Item	Supplier	Units	Unit Cost	Qty Each	Qty Needed	Extended Cost	Note
Chassis/Frame	Base Plate	Scrap Bin	ea	\$3.89	1	1	\$3.89	0.75 inch thick pine plywood
Mechatronics Kit	12v 100:1 Gear Motor	IDEA lab	ea	\$0.00	1	1	\$0.00	
Mechatronics Kit	12v 50:1 Gear Motor	IDEA lab	ea	\$0.00	1	1	\$0.00	
Mechatronics Kit	Small Solenoid	IDEA lab	ea	\$0.00	1	1	\$0.00	
Mechatronics Kit	Large Solenoid	IDEA lab	ea	\$0.00	1	1	\$0.00	
Mechatronics Kit	Pneumatic Valve	IDEA lab	ea	\$0.00	1	2	\$0.00	
Mechatronics Kit	Roller Limit Switch	IDEA lab	ea	\$0.00	1	1	\$0.00	
Mechatronics Kit	Long-Arm Limit Switch	IDEA lab	ea	\$0.00	1	1	\$0.00	
Mechatronics Kit	Pushbutton Switch	IDEA lab	ea	\$0.00	1	1	\$0.00	
Mechatronics Kit	Pneumatic Pressure Vessel Assembly	IDEA lab	ea	\$0.00	1	1	\$0.00	
Mechatronics Kit	6mm nylon tubing	IDEA lab	ea	\$0.00	1	1	\$0.00	
Mechatronics Kit	Arduino and Motor Driver Board	IDEA lab	ea	\$0.00	1	1	\$0.00	
Mechatronics Kit	12v 1.5A power supply	IDEA lab	ea	\$0.00	1	1	\$0.00	
Mechatronics Kit	12v 3A power supply	IDEA lab	ea	\$0.00	1	1	\$0.00	
Drive Train	Aluminum Axel	Amazon	ea	\$2.70	1	2	\$5.40	300 mm
Drive Train	606 Bearings	Amazon	ea	\$0.20	1	2	\$0.40	
Drive Train	6mm bore Flange Hub Coupler	Amazon	ea	\$1.67	1	4	\$6.68	
Drive Train	20 & 80 tooth timing belt reduction 6mm bore w/ belt	Amazon	ea	\$8.45	1	2	\$16.90	
Drive Train	Wheels (rear)	3d Printed	g	\$0.05	36	2	\$3.60	
Drive Train	Shaft Collar	3d Printed	g	\$0.05	3.99	4	\$0.80	
Drive Train	Side Walls	Home Depot	in^2	\$0.01	2	2	\$0.04	Laser-Cut 0.250" MDF
Drive Train	Front Wheels	Home Depot	ea	\$5.93	1	2	\$11.86	
Sheep Dispenser	Storage Columnn	3d Printed	g	\$0.05	175.42	1	\$8.77	
Sheep Dispenser	Platform	Scrap Bin	ea	\$0.10	1	1	\$0.10	
Sheep Dispenser	Collumns	Scrap Bin	ea	\$0.10	2	1	\$0.20	
Sweeping Mechanism	Wings	3d Printed	g	\$0.05	150	2	\$15.00	
Sweeping Mechanism	Pins	3d Printed	g	\$0.05	0.63	2	\$0.06	
Sweeping Mechanism	Collumns	Home Depot	in^2	\$0.01	4	2	\$0.08	
Extension	Telescoping arm Attachments	3d printed	g	\$0.05	2.6	1	\$0.13	
Side Walls	Foam Board	Michales	in^2	\$0.01	350	2	\$7.00	Decorative side walls (crafting foam board)
Extension	1/2" x 2' PVC Pipe	Home Depot	ea	\$3.98	1	1	\$3.98	
Extension	1" x 2' PVC Pipe	Home Depot	ea	\$4.86	1	1	\$4.86	
Extension	2" x 2' PVC Pipe	Scrap Bin	ea	\$5.00	1	1	\$5.00	
Extension	Cord Spool	3d Printed	g	\$0.05	47.56	1	\$2.38	
Fasteners	M6 screws	Home Depot	ea	\$0.09	1	51	\$4.59	
Fasteners	Bolts	Home Depot	ea	\$0.32	1	6	\$1.92	
Fasteners	Washers	Home Depot	ea	\$0.04	1	9	\$0.36	
Fasteners	Nuts	Home Depot	ea	\$0.03	1	6	\$0.18	
Fastener	Motor Mount (large)	3d Printed	ea	\$3.13	1	1	\$3.13	
Fastener	Motor Mount (small)	3d Printed	ea	\$3.13	1	1	\$3.13	
Fastener	Masking Tape Roll	Amazon	ea	\$2.99	1	1	\$2.99	
Fastener	Duct Tape Roll	Amazon	ea	\$4.48	1	1	\$4.48	
Fastener	Hot Glue Stick	Amazon	ea	\$1.00	1	1	\$1.00	
							\$118.91	

BOM
Total = \$118.91

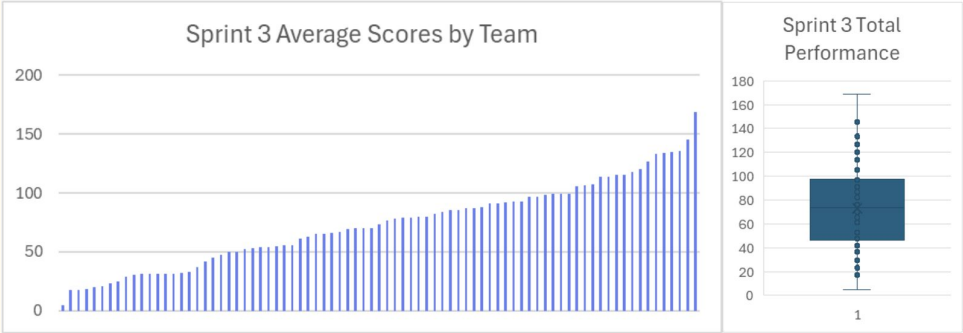
Final Results (Sprint 3 to Final Comp)

Sprint 3

Median: ~70 points

Our score: ~70 points

Next steps: finalize telescoping arm & continue extensive testing to ensure mechanism(s) success rates $\geq 95\%$



Final Comp.

43	A02	14	12.78571429	5.785714286	108.8571429	5.071428571	31.35714286	0	20
44	3	2	18.5	0	98.5	0	50	0	30
45	4	2	2.5	0	44	0	6.5	-2	15
46	5	3	23.33333333	9	151.6	23.66666667	50	2.666666667	30
47	6	2	0	0	18	0	6.5	0	2.5
48	1	3	22.33333333	18	122.25	0	50	0	30
49	2	2	0	0	18	0	6.5	-2	2.5
84	Grand Total	164	12.82317073	12.34756098	101.4632768	5.695121951	32.50609756	-0.432926829	25.12195122

Eliminated in the sweet 16 (T-16 result)

Overall; very satisfied with results considering our neglect of fish and eels task(s)